Dynamic Structural Selection

From Response Fields to Observable Cosmology in the MAAT Framework

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Abstract

Earlier papers in this series established structural selection as a ranking principle, defined the Critical Coherence Index (CCI), calibrated structural weights, and separated formal definitions from empirical proxies. This paper consolidates the dynamical extension of this programme: structural selection is promoted from a static ranking functional to a dynamical, local, gravitational, and observationally testable effective framework. This work establishes a continuous pipeline from structural diagnostics to observable cosmological signatures.

The core chain is

$$d_a \to C_{ab} \to \lambda_a(t) \to \lambda_a(x,t) \to T_{\mu\nu}^{\mathrm{MAAT}} \to \mathrm{FLRW} \to \mathrm{observables}.$$

Here d_a are primitive structural defects, C_{ab} is their covariance geometry, λ_a are response-generated selection pressures, and $T_{\mu\nu}^{\mathrm{MAAT}}$ is the effective stress–energy contribution obtained by metric variation of the MAAT action.

Three numerical benchmarks are reported. First, a covariance-response flow tracks a moving fixed point with final tracking error 9.91×10^{-7}. Second, local selection fields in a perturbed one-dimensional ϕ^4 kink benchmark reduce the final equation-residual ratio to 0.165 while leaving energy essentially unchanged. Third, a kinetic scalar FLRW scan finds 666/784 dynamically stable trajectories and 618/784 background-safe trajectories. Finally, the representative stable branch is converted into observable proxies, yielding $\Omega_{\mathrm{MAAT},0} = 0.00331$, $w_{\mathrm{MAAT},0} = -0.801$, maximum $|\Delta H/H| = 0.0399$, and maximum growth-proxy deviation $|\Delta f\sigma_8/f\sigma_8| = 0.0336$.

The result is not a completed fundamental theory and not a cosmological data fit. It establishes a reproducible effective-theory pipeline from structural diagnostics to observable signatures, while remaining a toy-level framework rather than a completed cosmological model.

1 Purpose

The first thirty papers of this series develop structural selection as an empirical and theoretical framework for nonlinear field systems, structural diagnostics, CCI observables, and response-based sector weights. However, a ranking functional alone is not yet a field theory.

The central consolidation step is to turn structural selection from a post-hoc ranking principle into an effective dynamical layer:

> *Selection pressures are not fixed parameters; they can be treated as response variables generated by defect covariance, promoted to local fields, coupled to gravity, and projected into observables.*

The paper is intentionally cautious. It does not claim a final theory of gravity, a fit to cosmological data, or a derivation of the Standard Model. It defines a closed effective chain and documents its first reproducible tests.

2 Version Map

Table 1: Consolidated version map.

Version	New layer	Role
v0.5	$\lambda_a(t)$	Structural weights become dynamical response variables.
v0.6	$\lambda_a(x,t)$	Response variables become local selection-pressure fields.
v0.7	$T_{\mu\nu}^{\mathrm{MAAT}}$	Local selection fields are coupled to gravity through metric variation.
v0.8	explicit scalar defect	A kinetic scalar defect is varied explicitly and stability conditions are identified.
v0.9	FLRW stability scan	The stable kinetic branch is tested in homogeneous cosmology.
v0.10	observable layer	The stable branch is projected into $H(z)$, $w(z)$, $\Omega(z)$, and growth proxies.

The resulting architecture is

$$\boxed{d_a \to C_{ab} \to \lambda_a(t) \to \lambda_a(x,t) \to S_{\mathrm{MAAT}} \to T_{\mu\nu}^{\mathrm{MAAT}} \to H(z),\ w(z),\ \Omega(z),\ f\sigma_8(z).} \tag{1}$$

3 Response Closure

Let $d_a[X]$ denote structural defect variables for sectors $a \in \{H, B, S, V, R\}$. The response-closed weight vector is defined by

$$\lambda^\star = \left(C + \eta \frac{\mathrm{tr}(C)}{A}\mathbb{I}\right)^{-1} \left(\bar{d} - d^\star\right), \tag{2}$$

where C is the defect covariance matrix, $\bar{d}$ is the reference-ensemble mean defect vector, $d^\star$ is the target defect vector, A is the number of structural sectors, and η is a ridge regularisation parameter.

This equation should be read as a linear-response closure. It does not make λ_a universal constants. It makes them effective response pressures associated with the covariance geometry of a chosen reference ensemble.

4 v0.5: Dynamic Selection Pressures

The static response closure becomes dynamic through relaxation to a moving response fixed point:

$$\boxed{\tau_\lambda \partial_t \lambda(t) = -\lambda(t) + \lambda^\star(t).} \tag{3}$$

Thus the earlier static closure is recovered as the fixed-point limit.

In the v0.5 benchmark, a correlated five-sector defect ensemble undergoes a temporary boundary-pressure event. The response weights adapt, with robustness pressure peaking during the event and relaxing afterward.

Table 2: v0.5 covariance-response flow.

Quantity	**Value**
Final loss	3.32624×10^{-3}
Minimum loss	2.50204×10^{-3}
Final $\|\lambda\|$	1.98115
Maximum λ_R	2.87546
Mean tracking error, last 100 steps	2.75463×10^{-5}
Final tracking error	9.90859×10^{-7}

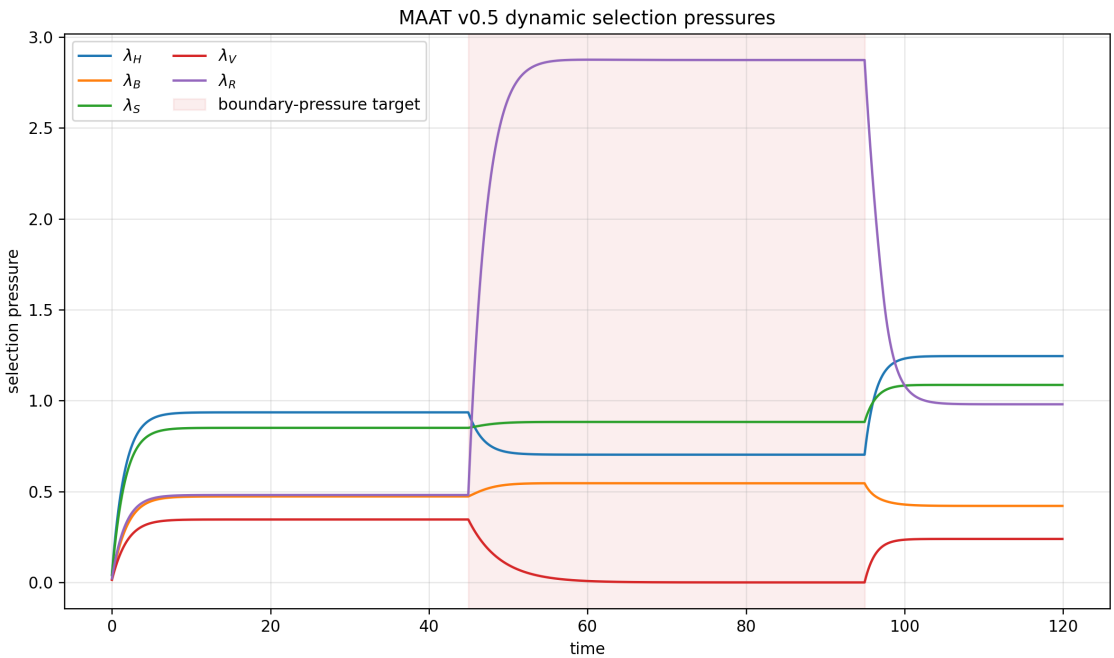

Figure 1: Dynamic response weights in the v0.5 benchmark. The boundary-pressure interval induces a temporary robustness-dominated response rather than a fixed hand-chosen weight hierarchy.

5 v0.6: Local Selection Fields

Local field theory requires local response variables:

$$\lambda_a(t) \longrightarrow \lambda_a(x,t). \tag{4}$$

The local response equation is

$$\boxed{\tau_a \partial_t \lambda_a(x,t) = D_a \nabla^2 \lambda_a(x,t) - \lambda_a(x,t) + \lambda_a^\star(x,t).} \tag{5}$$

The v0.6 numerical test uses a one-dimensional noisy ϕ^4 kink with local sectors (H, S, R). The local fields reduce equation-residual stress while keeping the final energy essentially unchanged.

Table 3: v0.6 local ϕ^4 benchmark. Ratios below one indicate improvement relative to baseline.

Metric	**MAAT / baseline**
Post-perturbation distance to clean kink	0.9682
Post-perturbation equation residual RMS	0.7576
Post-perturbation roughness	0.9995
Post-perturbation static energy	0.9975
Final equation residual RMS	0.1648
Final energy	1.00000000003

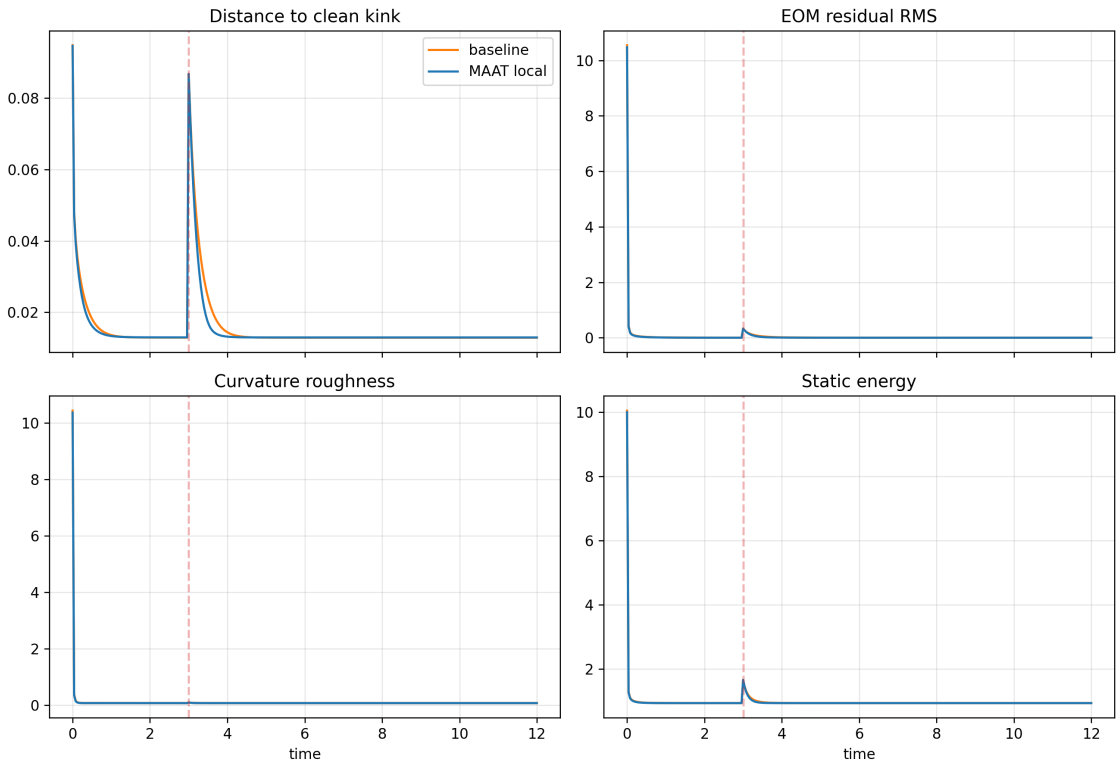

Figure 2: Local selection fields in the v0.6 one-dimensional ϕ^4 benchmark. The result is residual reduction without an energy-ranking reinterpretation.

6 v0.7 and v0.8: Gravitational Coupling

If $\lambda_a(x,t)$ are local fields, their gravitational role must be stated through an action and stress–energy tensor. The effective MAAT action is

$$S_{\mathrm{MAAT}} = \int \mathrm{d}^4x\,\sqrt{-g}\left[-M_*^4\sum_a \lambda_a \mathcal{U}_a - \frac{1}{2}\sum_a Z_a g^{\mu\nu}\nabla_\mu\lambda_a\nabla_\nu\lambda_a - V_\lambda(\lambda)\right], \tag{6}$$

with

$$\mathcal{U}_a = -\log(\epsilon + \Gamma_a), \qquad \Gamma_a = (1+d_a)^{-1}. \tag{7}$$

Metric variation defines

$$\boxed{T^{\mathrm{MAAT}}_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\mathrm{MAAT}}}{\delta g^{\mu\nu}}.} \tag{8}$$

The Einstein equation becomes

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N \left(T^{\text{matter}}_{\mu\nu} + T^{\text{MAAT}}_{\mu\nu} \right). \tag{9}$$

For a kinetic scalar defect,

$$X = -\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi, \qquad d_X = \frac{X}{\Lambda_X^4}, \tag{10}$$

one obtains

$$\frac{\partial d_X}{\partial g^{\mu\nu}} = -\frac{1}{2\Lambda_X^4} \nabla_\mu \phi \nabla_\nu \phi. \tag{11}$$

This is the first explicit worked example of the metric-variation term.

The important physical lesson is that the statistical sign of a support functional does not automatically fix the Lorentzian branch. The viable branch must satisfy

$$P_X > 0, \qquad c_s^2 = \frac{P_X}{P_X + 2X P_{XX}} > 0, \qquad \rho_{\text{MAAT}} \geq 0. \tag{12}$$

7 v0.9: FLRW Stability

The v0.9 benchmark scans the stable kinetic branch of a toy scalar model in homogeneous FLRW evolution over

$$\mu \in [10^{-3}, 10^3], \qquad \dot{\phi}_0 \in [0.03, 3.20], \tag{13}$$

on a 28×28 grid. It is a stability and consistency scan, not a complete cosmological analysis. The numerical count of stable trajectories is reported only as a phase-space outcome of this scan and carries no independent physical or symbolic interpretation.

Table 4: v0.9 FLRW stability scan.

Quantity	**Value**
Total trajectories	784
Dynamically stable trajectories	666
Background-safe trajectories	618
Stable fraction	0.8495
Background-safe fraction	0.7883
Representative $\max \Omega_{\text{MAAT}}$	0.02905
Representative $\min P_X$	1.36347
Representative $\max c_s^2$	1.82570

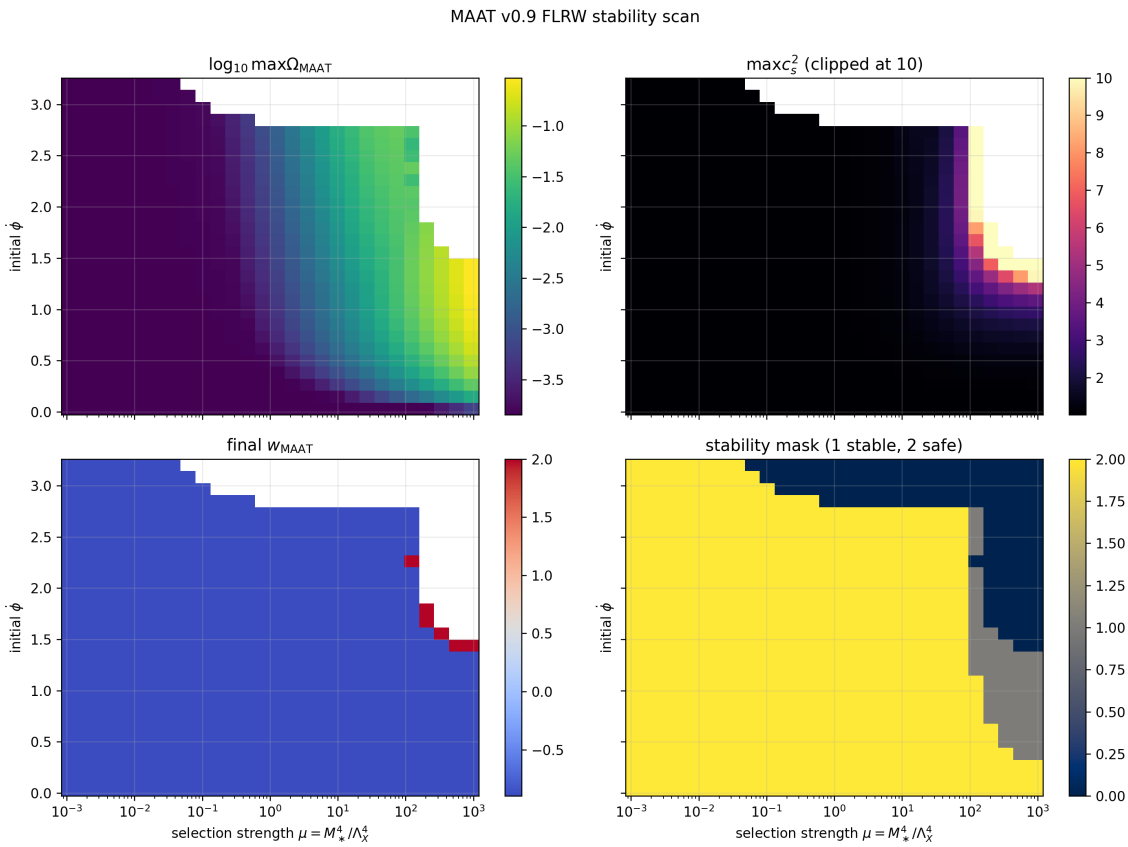

Figure 3: v0.9 FLRW stability scan. The benchmark identifies a broad stable region but also shows where gradient or background-safety failures occur.

8 v0.10: Observable Projection

Stability is not yet prediction. The v0.10 observable layer converts the representative stable v0.9 trajectory into expansion, equation-of-state, density-fraction, and growth proxies. The effective present epoch is chosen at $a \simeq 1$:

$$a_0 = 0.998777, \qquad H_0 = 1.002274, \tag{14}$$

and the redshift coordinate is

$$z = \frac{a_0}{a} - 1. \tag{15}$$

The internally normalised ΛCDM reference has

$$\Omega_{m0} = 0.300741, \tag{16}$$
$$\Omega_{r0} = 0.000100, \tag{17}$$
$$\Omega_{\Lambda 0} = 0.699158. \tag{18}$$

The Hubble deviation is

$$\frac{\Delta H}{H} = \frac{E_{\mathrm{MAAT}}(z) - E_{\Lambda\mathrm{CDM}}(z)}{E_{\Lambda\mathrm{CDM}}(z)}. \tag{19}$$

The growth proxy uses

$$f(z) \simeq \Omega_m(z)^{0.55}, \qquad f\sigma_8(z) = f(z)D(z)\sigma_{8,0}. \tag{20}$$

Table 5: v0.10 observable-projection summary.

Quantity	**Value**
Redshift range	$0 \leq z \leq 2.329$
$\Omega_{\mathrm{MAAT},0}$	0.00331
$w_{\mathrm{MAAT},0}$	-0.801
Maximum $\lvert\Delta H/H\rvert$	0.0399
Mean $\lvert\Delta H/H\rvert$	0.00481
Maximum $\lvert\Delta f\sigma_8/f\sigma_8\rvert$	0.0336

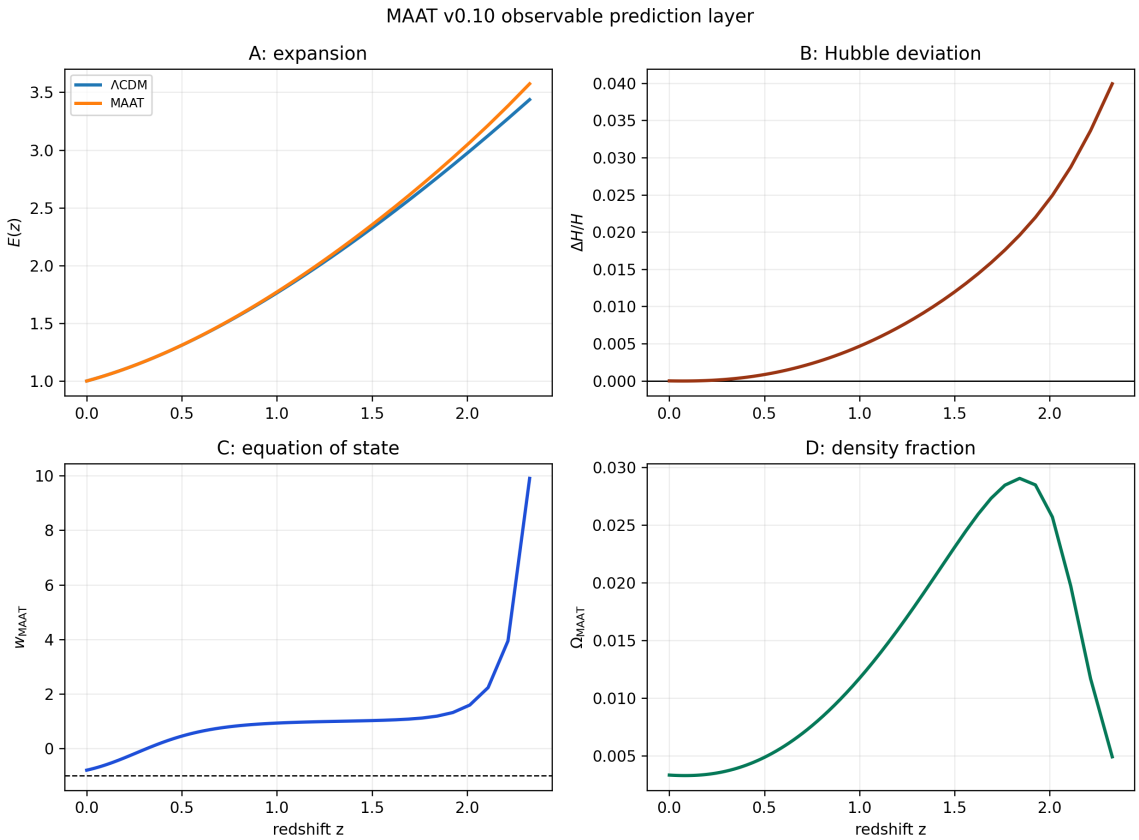

Figure 4: v0.10 observable layer. The representative stable branch is converted into expansion, Hubble deviation, equation-of-state, and density-fraction proxies.

9 Interpretation

The consolidated result is a ladder rather than a final theory:

$$\textit{ranking} \to \textit{response} \to \textit{local fields} \to \textit{gravity} \to \textit{stability} \to \textit{observables}.$$

The most important conceptual shift is that structural selection is no longer only a score assigned after the fact. It becomes an adaptive pressure generated by the covariance geometry of defects. In local form, that pressure behaves as a field. If covariantly defined, it contributes an effective stress–energy tensor. In this sense, MAAT acts as an effective selection layer modifying cosmological dynamics without replacing the underlying dynamical theory.

Within the v0.10 toy projection, the model predicts small but structured deviations from ΛCDM in expansion and approximate growth observables. These deviations are not yet data-level predictions, but they define the quantities that a future observational test must confront.

Their percent-level scale is, in principle, the kind of regime targeted by precision expansion and growth surveys such as DESI, Euclid, CMB-S4, and next-generation weak-lensing analyses, once a full perturbation calculation is available.

At the same time, the limitations are central. The v0.9 and v0.10 tests are toy-model diagnostics. They are not observational fits, they do not replace ΛCDM, and they do not establish a complete modified-gravity theory. Their value is that they define what must be tested next.

10 Reproducibility

All numerical components are stored in the Structural Selection repository:

https://github.com/Chris4081/structural-selection-principle

The relevant experiment folders are:

- `experiments/maat_dynamic_fields_v05_v09/`
- `experiments/maat_observable_predictions_v10/`

The main reproduction commands are:

```
cd experiments/maat_dynamic_fields_v05_v09/v05_dynamic_lambda_flow
python3 lambda_dynamic_flow_v05.py

cd ../v06_local_selection_fields
python3 local_selection_phi4_v06.py

cd ../v09_flrw_stability_scan
python3 maat_flrw_stability_scan_v09.py

cd ../../maat_observable_predictions_v10
python3 maat_observable_predictions_v10.py
```

Each script writes CSV/JSON outputs and figures to a local `outputs/` directory.

11 Open Problems

Several issues remain open:

- The full five-sector covariant defect basis has not yet been varied.
- The local response-field equations need an open-system derivation connecting dissipative relaxation to covariant dynamics.
- The cosmological branch has not been fit to real data.
- A full perturbation solver is needed before the growth proxy can be treated as an observational prediction.
- Fifth-force, gravitational-wave, CMB, BAO, supernova, and weak-lensing constraints are not yet included.

12 Conclusion

Paper 31 consolidates the MAAT versioned development into a single effective framework. The sequence v0.5–v0.10 shows how structural selection can be made dynamic, local, gravitationally coupled, stable in a toy FLRW setting, and projected into observable signatures. It therefore provides a single reproducible pipeline from structural diagnostics to observable cosmological signatures.

The strongest result is not a claim of completeness. It is operational closure: the framework now produces equations, simulations, figures, and observable proxies that can be reproduced and challenged.

The next hard step is a genuine cosmological confrontation: implement perturbations, compute likelihoods, and test whether the percent-level observable deviations survive real data.

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